

Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.61_e-1)

Metric

$$ds^2 = -\frac{x^2}{N^2}dt \otimes dt - \frac{n}{N^2}dt \otimes d\varphi - \frac{x^2}{(x^4 - bx^2 \log(x) - cx^2 - n^2)N^2}dx \otimes dx - \frac{n}{N^2}d\varphi \otimes dt + \left(-\frac{x^2 - b \log(x) - c}{N^2}\right)d\varphi \otimes d\varphi + \frac{1}{N^2}dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, \varphi, z)$$

Metric Functions

$$N(z) = N$$

$$G(x) = bx^2 \log(x) - x^2 + c + \frac{n^2}{x^2}$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}x}{2N}\right) dt + \left(-\frac{\sqrt{2}\sqrt{-x^2 + b \log(x) + c + \frac{n^2}{x^2}}}{2N} + \frac{\sqrt{2}n}{2xN}\right) d\varphi$$

$$n = \left(\frac{\sqrt{2}x}{2N}\right) dt + \left(\frac{\sqrt{2}\sqrt{-x^2 + b \log(x) + c + \frac{n^2}{x^2}}}{2N} + \frac{\sqrt{2}n}{2xN}\right) d\varphi$$

$$m = \left(\frac{1}{\sqrt{-2x^2 + 2b \log(x) + 2c + \frac{2n^2}{x^2}N}}\right) dx + \left(\frac{i\sqrt{2}}{2N}\right) dz$$

$$\bar{m} = \left(\frac{1}{\sqrt{-2x^2 + 2b \log(x) + 2c + \frac{2n^2}{x^2}N}} \right) dx + \left(-\frac{i\sqrt{2}}{2N} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}x^4N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}bx^2N \log(x)}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}cx^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}n^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \right. \\ \left. - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2nN}}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \right) dt + \left(\frac{\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2xN}}{\sqrt{2}x^4 - \sqrt{2}bx^2 \log(x) - \sqrt{2}cx^2 - \sqrt{2}n^2} \right) d\varphi$$

$$n = \left(-\frac{\sqrt{2}x^4N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}bx^2N \log(x)}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}cx^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}n^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \right. \\ \left. + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2nN}}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \right) dt + \left(-\frac{\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2xN}}{\sqrt{2}x^4 - \sqrt{2}bx^2 \log(x) - \sqrt{2}cx^2 - \sqrt{2}n^2} \right) d\varphi$$

$$m = \left(\frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2N}}{2x} \right) dx + \left(\frac{1}{2}i\sqrt{2}N \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2N}}{2x} \right) dx + \left(-\frac{1}{2}i\sqrt{2}N \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}x^4N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}bx^2N \log(x)}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}cx^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \\ + \frac{\sqrt{2}n^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2nN}}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \partial_t + \frac{\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2xN}}{\sqrt{2}x^4 - \sqrt{2}bx^2 \log(x) - \sqrt{2}cx^2 - \sqrt{2}n^2} \partial_\varphi$$

$$\Delta = -\frac{\sqrt{2}x^4N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}bx^2N \log(x)}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}cx^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \\ + \frac{\sqrt{2}n^2N}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2nN}}{2(x^5 - bx^3 \log(x) - cx^3 - n^2x)} \partial_t - \frac{\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2xN}}{\sqrt{2}x^4 - \sqrt{2}bx^2 \log(x) - \sqrt{2}cx^2 - \sqrt{2}n^2} \partial_\varphi$$

$$\delta = \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2x} \partial_x + \frac{1}{2}i\sqrt{2}N \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}N}{2x} \partial_x - \frac{1}{2}i\sqrt{2}N \partial_z$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned} \kappa = & \frac{\sqrt{2}nx^4N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}bnx^2N \log(x)}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2N \log(x)}{4(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}cnx^2N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} \\ & - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2N}{8(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}cx^2N}{4(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}n^3N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}n^2N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} \end{aligned}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{\sqrt{2}nx^4N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}bnx^2N \log(x)}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2N \log(x)}{4(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}cnx^2N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} \\ & + \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}bx^2N}{8(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}cx^2N}{4(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}n^3N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2 \log(x) + cx^2 + n^2}n^2N}{2(x^6 - bx^4 \log(x) - cx^4 - n^2x^2)} \end{aligned}$$

$$\alpha = -\frac{1}{4}i\sqrt{2}\frac{\partial}{\partial z}N + \frac{\sqrt{2}nN}{4x^2}$$

$$\beta = -\frac{1}{4}i\sqrt{2}\frac{\partial}{\partial z}N + \frac{\sqrt{2}nN}{4x^2}$$

$$\begin{aligned}\tau = & \frac{i\sqrt{2}x^4\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}bx^2\log(x)\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}cx^2\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}x^2N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} \\ & + \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}bN\log(x)}{4(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}n^2\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}bN}{8(x^4 - bx^2\log(x) - cx^2 - n^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}cN}{4(x^4 - bx^2\log(x) - cx^2 - n^2)}\end{aligned}$$

$$\begin{aligned}\pi = & \frac{i\sqrt{2}x^4\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}bx^2\log(x)\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}cx^2\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} + \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}x^2N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} \\ & - \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}bN\log(x)}{4(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{i\sqrt{2}n^2\frac{\partial}{\partial z}N}{2(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}bN}{8(x^4 - bx^2\log(x) - cx^2 - n^2)} - \frac{\sqrt{2}\sqrt{-x^4 + bx^2\log(x) + cx^2 + n^2}cN}{4(x^4 - bx^2\log(x) - cx^2 - n^2)}\end{aligned}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{bN^2}{4x^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{1}{2}N^2 - \frac{1}{2}N\frac{\partial^2}{(\partial z)^2}N$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{4}N^2 + \frac{1}{4}N\frac{\partial^2}{(\partial z)^2}N + \frac{bN^2}{8x^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{1}{2} N^2 - \frac{1}{2} N \frac{\partial^2}{(\partial z)^2} N$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{bN^2}{4x^2}$$

$$\Lambda = \frac{1}{4} N^2 - \frac{1}{2} \frac{\partial}{\partial z} N^2 + \frac{1}{4} N \frac{\partial^2}{(\partial z)^2} N - \frac{bN^2}{24x^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{bN^2}{4x^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{bN^2}{12x^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{bN^2}{4x^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*D*"